YeongUk Choi

M.S. & Ph.D. Candidate
Department of Physics, Inha University · QFML & NMPL

aowlr8945@gmail.com yeonguk-choi.github.io Google Scholar



RESEARCH FOCUS

Oxygen-vacancy-gradient VO_2
Studying how spatial oxygen-vacancy gradients reshape electrical transport in vanadium dioxide.

EDUCATION

2024.03 – present
Integrated M.S.–Ph.D. Program, Physics
Inha University, Incheon, Republic of Korea
Quantum Functional Materials Lab (QFML) · Nano Materials Physics Lab (NMPL)

2018.03 – 2024.03
B.S. in Physics
Inha University, Incheon, Republic of Korea

AWARDS

2025.04
Outstanding Presentation Award — Poster Session
The Korean Physical Society (KPS) · 2025 KPS Spring Meeting
"Investigation of phase transition in self-flux grown VO_2 single crystals"

LABORATORY SKILLS

Measurement
PPMS (1.8 – 400 K · ±9 T) — R(T) · R(H) · Hall effect · VSM magnetometry · Heat capacity · Angle-dependent transport (rotator)
Synchrotron methods (beamline) — X-ray diffraction (XRD) · X-ray photoelectron spectroscopy (XPS) · X-ray absorption spectroscopy (XAS)
Specialized transport (low T) — Andreev spectroscopy · Gate-dependent transport

Growth
Pulsed-laser deposition (PLD) (Nd:YAG · 266 nm) — Epitaxial oxide, chalcogenide and intermetallic thin films

Analysis & software
Modeling & analysis (COMSOL · Rietveld) — COMSOL Multiphysics simulation · Rietveld refinement of diffraction data

Materials grown
CrO_2 · $LaNiO_3$ · RuO_2 · $SrFeO_3$ · VO_2 · W:VO_2 · LSMO · LCMO · Cu_2S · FeSe · $NbSe_2$ · Nb_3Sn · $Ni_{0.8}Fe_{0.2}$ · NiMnSb

PUBLICATIONS · 11 PEER-REVIEWED

1. **Strain Effect on Metal–Insulator Transition and Electronic Structure of Epitaxial VO_2 Thin Films: Comparative Structural, Transport, and Optical Studies** SELECTED
   Kyung Hoon Kim, **Yeong Uk Choi**, Ju Won Jeong, Joo Hyeong Lee, Hyun Soo Ahn, Yuna Lee, Ki-Ju Yee, Kyu-Tae Lee, Jong Hun Kim, Jung-Woo Lee, Jong Hoon Jung
   physica status solidi (RRL) – Rapid Research Letters 20 e70202 (2026) · doi:10.1002/pssr.70202

2. **Investigation of phase transition in self-flux grown VO_2 single crystals** SELECTED
Yeong Uk Choi, Joo Chan Kim, Jong Hun Kim, Jong Hoon Jung
Journal of the Korean Physical Society 86 800-805 (2025) · doi:10.1007/s40042-025-01330-9

3. **Specific heat investigation of superconducting properties in 2H-$NbSe_2$ single crystals** SELECTED
Joo Chan Kim, **Yeong Uk Choi**, Su Min Lee, Jong Yeon Kim, Jung Jin Park, Min Su Yeo, Jong Hun Kim, Jong Hoon Jung
Journal of the Korean Physical Society 89 174-181 (2026) · doi:10.1007/s40042-026-01632-6

4. **Promoting Gas Sensitivity of Graphitic Carbon Nitride via Incorporation of Platinum Nanoparticles under Blue-Light Irradiation**
Jong Hun Kim, Gayoung Yoon, Dheeraj Kumar, **Yeong Uk Choi**, Ying Chieh Hu, Ali Mirzaei, Kyuwook Ihm, Jeong Young Park, Sang Sub Kim, Jong Hoon Jung, Jae-Hun Kim
ACS Sensors acssensors.6c01299 (2026) · doi:10.1021/acssensors.6c01299

5. **Phase transitions, dielectric response and lattice dynamics of dimethylammonium mixed lead halide perovskites**
Furqanul Hassan Naqvi, Syed Bilal Junaid, Jae-Hyeon Ko, Hyun Jung Kim, Hyoungjeen Jeen, Wonhyuk Shon, Seongsu Lee, Seong Heon Kim, **Yeong Uk Choi**, Jong Hoon Jung
Materials Today Physics 61 102024 (2026) · doi:10.1016/j.mtphys.2026.102024

6. **Form Factor–Free Magneto-Triboelectric Generator for Standalone Power Line IoT Applications**
Hyun Soo Kim, Min Hyuk Lee, Do-Heon Kim, Dong-Gyu Lee, Iman M. Imani, SungHoon Hur, Young Joon Ko, **Yeong Uk Choi**, Hyunah Cho, So-Min Song, Tae Kyoung Yoon, In Woo Oh, Jong Hoon Jung, Jun Chen, Yunseok Kim, Heemin Kang, Jungho Ryu, Jeong Min Baik, Hyun-Cheol Song
Advanced Energy Materials 15 2500856 (2025) · doi:10.1002/aenm.202500856

7. **Suppression of Phase Transition and Dipolar Glassy Behavior in Dimethylammonium/Methylammonium Lead Chloride Single Crystals**
Syed Bilal Junaid, Furqanul Hassan Naqvi, Jae-Hyeon Ko, Seongsu Lee, Wonhyuk Shon, Joo Chan Kim, **Yeong Uk Choi**, Jong Hoon Jung
ACS Applied Energy Materials 8 9140-9155 (2025) · doi:10.1021/acsaem.5c00786

8. **Dynamics of glassy state in $MAPbBr_{3-x}Cl_x$ mixed lead halide perovskites**
Furqanul Hassan Naqvi, Syed Bilal Junaid, Jae-Hyeon Ko, **Yeong Uk Choi**, Jong Hoon Jung, Wonhyuk Shon, Seongsu Lee
Ceramics International 50 40136-40150 (2024) · doi:10.1016/j.ceramint.2024.07.399

9. **Strain-insensitive ferromagnetic $SrRuO_3$ thin films with ferrimagnetic $CoFe_2O_4$ buffer layer**
Jung Ehy Hong, **Yeong Uk Choi**, Hyun Soo Ahn, Bhubnesh Lama, Jong Hun Kim, Tula R. Paudel, Jung-Woo Lee, Jong Hoon Jung
Current Applied Physics 66 24-29 (2024) · doi:10.1016/j.cap.2024.06.012

10. **$CoFe_2O_4$ on Mica Substrate as Flexible Ethanol Gas Sensor in Self-Heating Mode**
Jong Hun Kim, **Yeong Uk Choi**, Jong Hoon Jung, Jae-Hun Kim
Sensors 24 1927 (2024) · doi:10.3390/s24061927

11. **Optimization of a $LaNiO_3$ Bottom Electrode for Flexible $Pb(Zr,Ti)O_3$ Film-Based Ferroelectric Random Access Memory Applications**
Yeong Uk Choi, Hyun Soo Ahn, Jung Ehy Hong, Dong In Yang, Hwa-Pyeong Lee, Dae-Yong Jeong, Minbaek Lee, Jong Hun Kim, Jong Hoon Jung
Crystals 13 1613 (2023) · doi:10.3390/cryst13121613